

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083 - NetworkWalks
Date	18 September 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) & W2-PM5 (Zenmap Scanning)
Target Domain	networkwalks.com
Target Subnet	10.42.27.0/24

1. Project Title

Footprinting and Scanning using Kali Linux and Zenmap

2. Project Description

This project is part of an ethical hacking lab exercise. The main objective is to perform reconnaissance and scanning tasks to collect evidence about a target domain and subnet. Reconnaissance (information gathering) is the first step in penetration testing, while scanning helps identify live hosts and network topology.

The project is divided into two modules:

- Elective Module (Kali Linux Tools) → WHOIS, WhatWeb, Nslookup, Curl, WAFW00F, DNSRecon.
- Essential Module (Zenmap Scanning) → Ping Scan on subnet 10.42.27.0/24, host discovery, IP/MAC collection, and topology visualization.

3. Objectives

- Gather domain registration details.
- Detect web technologies and DNS records.
- Identify firewall presence.
- Discover live hosts in subnet.
- Collect IP and MAC addresses of hosts.
- Generate and save network topology PDF.
- Document all findings in a structured final report.

4. Methodology

4.1 Elective Module (Kali Linux) — W2-PM1

Ran WHOIS, WhatWeb, Nslookup, Curl, WAFW00F, and DNSRecon against the target domain networkwalks.com. Each output was saved into a .txt file for evidence.

4.2 Essential Module (Zenmap) — W2-PM5

1. Task 1: Ran Ping Scan on subnet 10.42.27.0/24 using Zenmap.
2. Task 2: Identified live hosts (IP + MAC addresses).
3. Task 3: Generated topology view of the subnet.
4. Task 4: Counted total live hosts in the subnet.
5. Task 5: Listed all IP addresses of live hosts.
6. Task 6: Listed all MAC addresses of live hosts.
7. Task 7: Saved topology output as a PDF file with legend enabled.

4.3 Evidence Collection

Outputs were organized into folders for clarity:

- Module1_Elective → Kali Linux .txt outputs (WHOIS, WhatWeb, Nslookup, Curl, WAFW00F, DNSRecon)
- Module1_Essential → Zenmap .txt output + topology PDF

A final consolidated report (W2-PM-FINAL) was then prepared covering both modules.

5. Elective Module Evidence (W2-PM1 — Kali Linux Tools)

Task 1: WHOIS — Domain Registration Details

Command used:

```
$ whois networkwalks.com
```

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-18T14:50:29Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to  
use electronic processes that are automated and high-volume to access or  
query the Whois database except as reasonably necessary to register
```

Finding: WHOIS revealed the domain registrar, registration/expiry dates, and name servers, which point to the hosting provider.

Task 2: WhatWeb — Web Technology Fingerprinting

Command used:

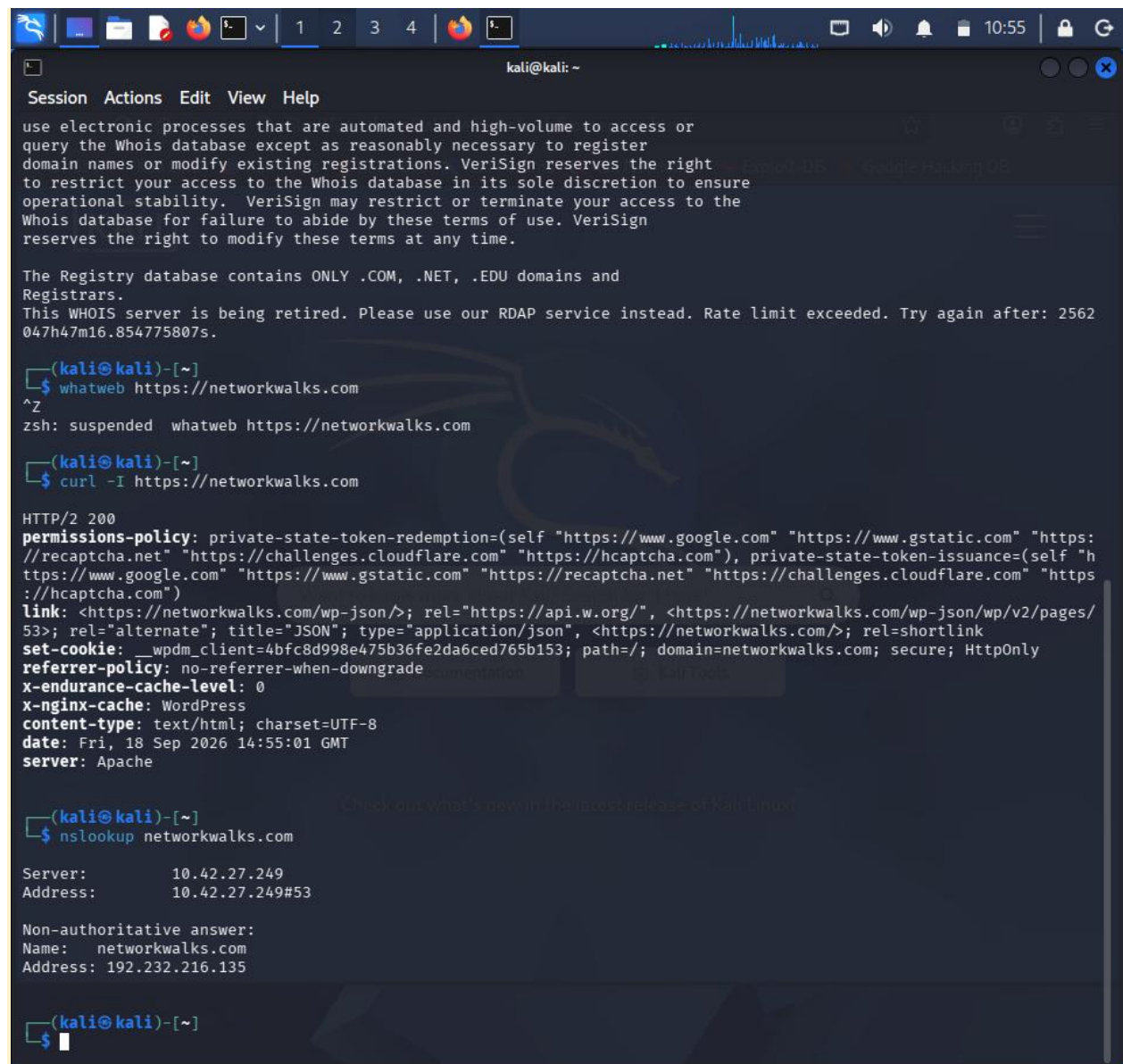
```
$ whatweb networkwalks.com
```

Finding: WhatWeb exposed the exact CMS, plugin versions, server IP, and an associated email address.

Task 3: Nslookup — DNS Resolution

Command used:

```
$ nslookup networkwalks.com
```



```
kali@kali: ~  
Session Actions Edit View Help  
use electronic processes that are automated and high-volume to access or  
query the Whois database except as reasonably necessary to register  
domain names or modify existing registrations. VeriSign reserves the right  
to restrict your access to the Whois database in its sole discretion to ensure  
operational stability. VeriSign may restrict or terminate your access to the  
Whois database for failure to abide by these terms of use. VeriSign  
reserves the right to modify these terms at any time.  
  
The Registry database contains ONLY .COM, .NET, .EDU domains and  
Registrars.  
This WHOIS server is being retired. Please use our RDAP service instead. Rate limit exceeded. Try again after: 2562  
047h47m16.854775807s.  
  
(kali@kali)-[~]  
$ whatweb https://networkwalks.com  
^Z  
zsh: suspended whatweb https://networkwalks.com  
  
(kali@kali)-[~]  
$ curl -I https://networkwalks.com  
  
HTTP/2 200  
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink  
set-cookie: _wpdm_client=4bfc8d998e475b36fe2da6ced765b153; path=/; domain=networkwalks.com; secure; HttpOnly  
referrer-policy: no-referrer-when-downgrade  
x-endurance-cache-level: 0  
x-nginx-cache: WordPress  
content-type: text/html; charset=UTF-8  
date: Fri, 18 Sep 2026 14:55:01 GMT  
server: Apache  
  
(kali@kali)-[~]  
$ nslookup networkwalks.com  
  
Server:          10.42.27.249  
Address:         10.42.27.249#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135  
  
(kali@kali)-[~]  
$
```

Finding: Nslookup resolved the domain to its real IP address, enabling direct server-level scanning.

Task 4: Curl — HTTP Response Headers

Command used:

```
$ curl -I https://networkwalks.com
```

```
Kali Linux — Mozilla Firefox
Session Actions Edit View Help

information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
This WHOIS server is being retired. Please use our RDAP service instead. Rate limit exceeded. Try again after: 2562
047h47m16.854775807s.

(kali@kali)-[~]
$ whatweb https://networkwalks.com
^Z
zsh: suspended  whatweb https://networkwalks.com

(kali@kali)-[~]
$ curl -I https://networkwalks.com

HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=4bfc8d998e475b36fe2da6ced765b153; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 18 Sep 2026 14:55:01 GMT
server: Apache

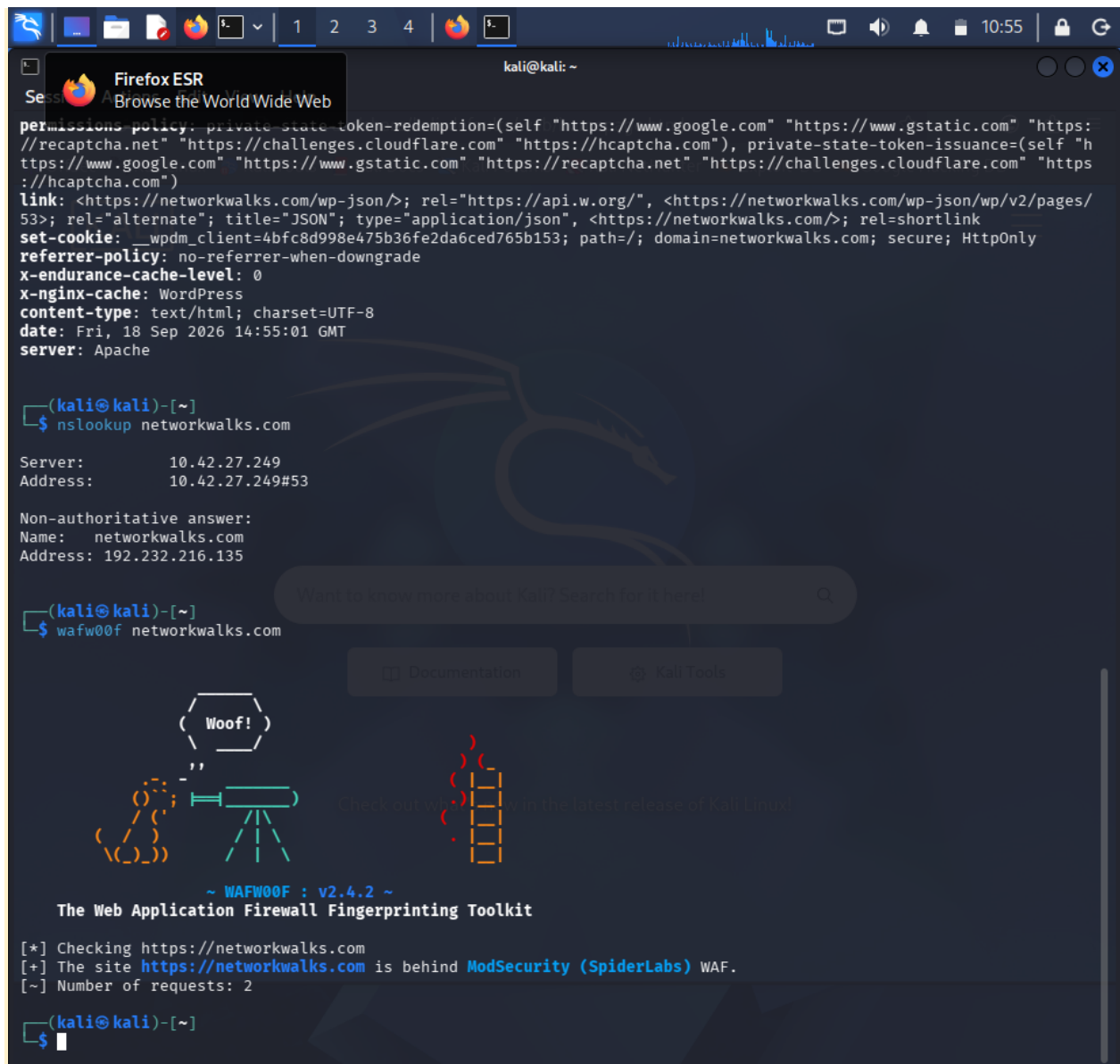
(kali@kali)-[~]
$
```

Finding: HTTP headers revealed the web server, caching layer, and hidden API endpoints.

Task 5: WAFW00F — Web Application Firewall Detection

Command used:

```
$ wafw00f networkwalks.com
```



Finding: The target site was found to be protected by a Web Application Firewall (WAF).

Task 6: DNSRecon — DNS Record Enumeration

Command used:

```
$ dnsrecon -d networkwalks.com
```

```
kali@kali: ~  
[+] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2  
  
-(kali@kali)-[~]  
$ dnsrecon -d networkwalks.com  
  
2026-09-18T10:56:21.137775-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-09-18T10:56:21.140581-0400 INFO std: Performing General Enumeration against: networkwalks.com ...  
2026-09-18T10:56:21.824837-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-09-18T10:56:22.242205-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-09-18T10:56:22.954181-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-09-18T10:56:23.635875-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-09-18T10:56:23.636310-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-09-18T10:56:24.227331-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-09-18T10:56:24.400132-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-09-18T10:56:24.416463-0400 INFO A networkwalks.com 192.232.216.135  
2026-09-18T10:56:25.075327-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X  
-Ij-mjgEeg-UsYI  
2026-09-18T10:56:25.075770-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelco  
me.com ~all  
2026-09-18T10:56:26.308411-0400 INFO Enumerating SRV Records  
2026-09-18T10:56:36.141000-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.14 443  
2026-09-18T10:56:36.141659-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.15 443  
2026-09-18T10:56:36.142534-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.11 443  
2026-09-18T10:56:36.142683-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.15 443  
2026-09-18T10:56:36.143032-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.9 443  
2026-09-18T10:56:36.143159-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.11 443  
2026-09-18T10:56:36.143249-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.9 443  
2026-09-18T10:56:36.143354-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.14 443  
2026-09-18T10:56:36.147346-0400 INFO 8 Records Found  
2026-09-18T10:56:36.147757-0400 INFO Completed enumeration for domain: networkwalks.com  
  
-(kali@kali)-[~]  
$
```

Finding: DNSRecon enumerated all DNS records including name servers, mail servers, SPF/TXT records, and service (SRV) records.

6. Essential Module Evidence (W2-PM5 — Zenmap Scanning)

Task 1 Download & install Zenmap from the official website.

[Paste screenshot of Zenmap installed / opened]

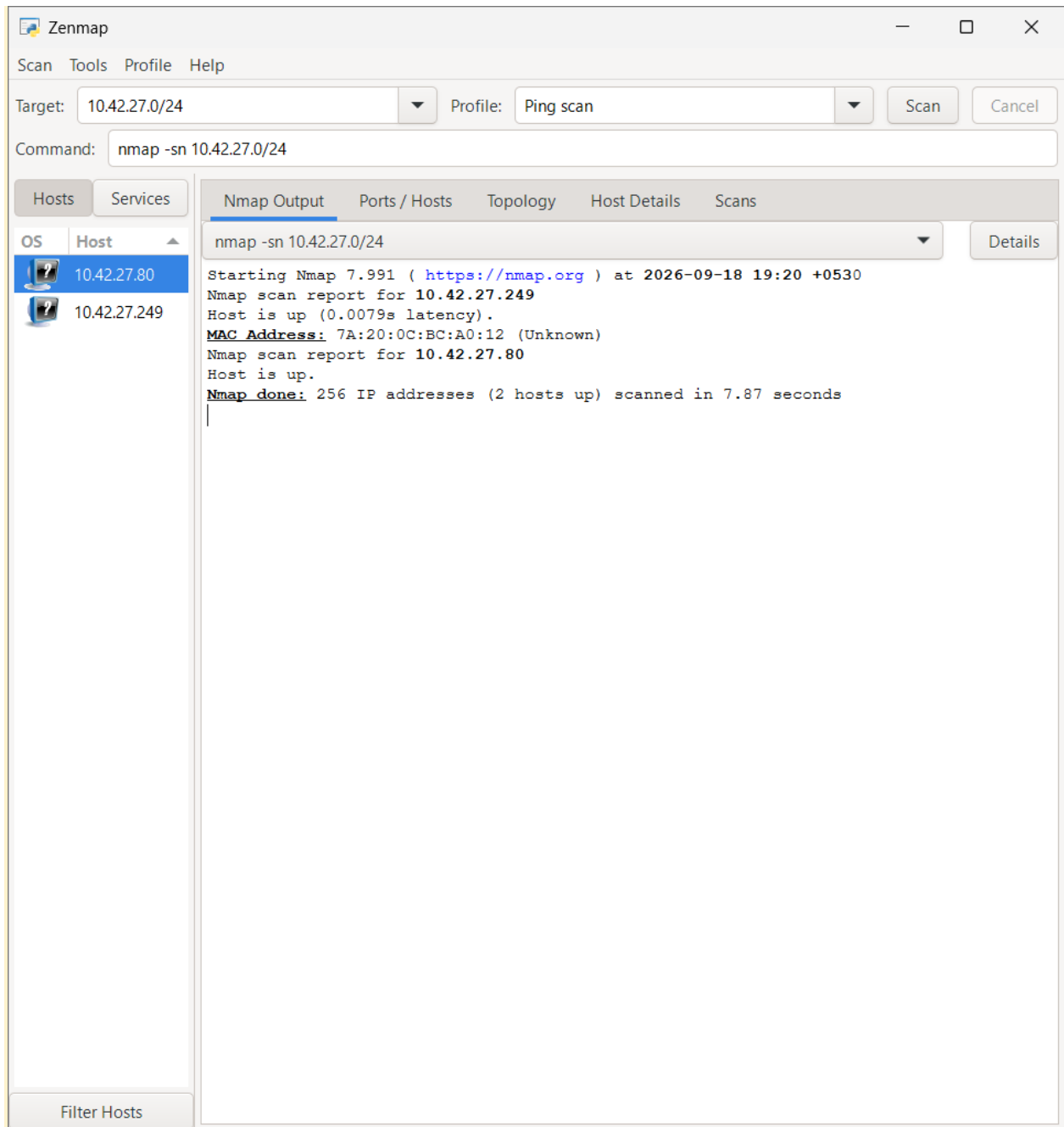
Task 2 Find local IP address & LAN subnet using ipconfig.

[Paste screenshot of ipconfig output]

Task 3 Find the list of live hosts/PCs in the IP subnet (Ping Scan).

Command used:

```
nmap -sn 10.42.27.0/24
```



Task 4 How many hosts are live in the subnet?

[Answer: [X] hosts are live (including my PC)]

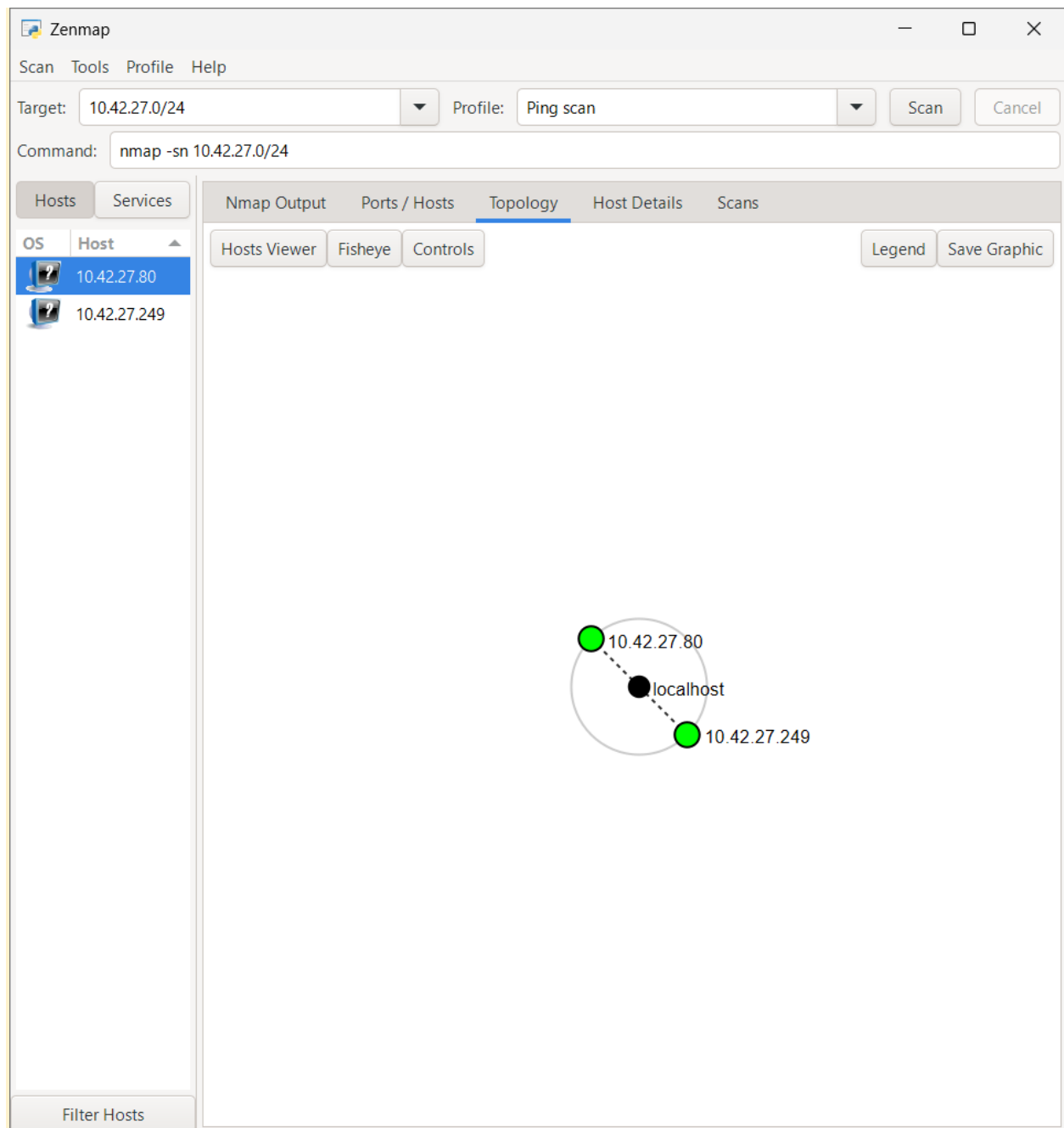
Task 5 What are the IP addresses of the live hosts?

[List the IP addresses found in your scan, e.g. 10.42.27.1, 10.42.27.4, ...]

Task 6 What are the MAC addresses of the live hosts?

[List the MAC addresses found in your scan]

Task 7 Display & save the output topology in PDF format.



Finding: The Zenmap Ping Scan discovered all active hosts within the target subnet 10.42.27.0/24, and the topology view visualized their relationship to the local machine.

7. Results

- WHOIS revealed domain registration details.
- WhatWeb detected web technologies in use.
- Nslookup and DNSRecon enumerated DNS records.
- WAFW00F identified firewall presence protecting the target.
- Curl displayed HTTP response headers.
- Zenmap scanning discovered live hosts in subnet 10.42.27.0/24.
- IP and MAC addresses of all live hosts were collected.
- A topology PDF was generated to visualize the network structure.

8. Conclusion

The project successfully demonstrated reconnaissance and scanning techniques using Kali Linux and Zenmap. Evidence was collected, organized, and documented in this final report. This exercise provided practical, hands-on experience in ethical hacking fundamentals and network analysis, forming a strong foundation for later stages of the penetration testing lifecycle.

9. Disclaimer

This report was prepared strictly for educational and research purposes as part of the NetworkWalks Cybersecurity Internship Program. All reconnaissance and scanning activities were performed against a permitted lab target / self-owned environment. No unauthorized access or testing was performed against any external system without proper authorization.